

WTW 158 - MEMO - **WORKSHEET ON DIFFERENTIATION -2013**

Find the derivatives of the following functions.

1. $\frac{d}{dx} x^2 e^{4x} = 2x \times e^{4x} + x^2 \times 4e^{4x}$
2. $\frac{d}{d\theta} \frac{\tan 2\theta}{1 + \sec(\theta^2)} = \frac{2 \sec^2 2\theta [1 + \sec(\theta^2)] - \tan 2\theta \sec(\theta^2) \tan(\theta^2) 2\theta}{[1 + \sec(\theta^2)]^2}$
3. $\frac{d}{dt} (t^3 + e^t + 3^t) = 3t^2 + e^t + 3^t \ln 3$
4. $\frac{d}{dt} [(1 - 4t)^3 + e^{1-5t} + 3^{1-6t}] = 3(1 - 4t)^2(-4) + e^{1-5t}(-5) + (\ln 3)3^{1-6t}(-6)$
5. $\frac{d}{dx} 5 \sin(x^4) = 5 \times \cos(x^4) \times 4x^3$
6. $\frac{d}{dx} \sin^4 5x = 4 \sin^3(5x) \times \cos(5x) \times 5$
7. $\frac{d}{dt} e^{\cos 5t} = e^{\cos 5t} \times -\sin 5t \times 5$
8. $\frac{d}{dt} \cos(e^{5t}) = -\sin(e^{5t}) \times e^{5t} \times 5$
9. $\frac{d}{du} \ln(1 - \sqrt[3]{u}) = \frac{1}{1 - \sqrt[3]{u}} \times -\frac{1}{3} u^{-2/3}$
10. $\frac{d}{du} \sqrt[3]{1 - \ln 2u} = \frac{1}{3} (1 - \ln 2u)^{-2/3} \times \frac{2}{2u}$
11. $\frac{d}{dx} \sec(\ln(2x + 6)) = \sec(\ln(2x + 6)) \tan(\ln(2x + 6)) \times \frac{2}{2x + 6}$
12. $\frac{d}{dt} \ln(\sec(3x + 7)) = \frac{1}{\sec(3x + 7)} \times \sec(3x + 7) \tan(3x + 7) \times 3$
13. $\frac{d}{dx} \operatorname{cosec}(3x^2) = -\operatorname{cosec}(3x^2) \cot(3x^2) \times 6x$
14. $\frac{d}{dx} 3^{\operatorname{cosec} x^2} = (\ln 3) 3^{\operatorname{cosec} x^2} \times -\operatorname{cosec}(x^2) \cot(x^2) \times 2x$
15. $\frac{d}{dx} \tan^{-1}(e^{8x}) = \frac{8e^{8x}}{1 + (e^{8x})^2}$
16. $\frac{d}{dt} \sin^{-1}(1 - 9t^4) = \frac{1}{\sqrt{1 - (1 - 9t^4)^2}} \times -36t^3$
17. $x^3 \sin y + xy^4 = y + e \Rightarrow$
 $3x^2 \sin(y) + x^3 \cos(y)y' + y^4 + 4xy^3y' = y' \Rightarrow y' = -\frac{3x^2 \sin y + y^4}{x^3 \cos y + 4xy^3 - 1}$

$$18. f(t) = t^{\cos 2t} \Rightarrow \ln f(t) = \ln t^{\cos 2t} = \cos 2t \ln t$$

$$\Rightarrow \frac{d}{dt} \ln f(t) = \frac{d}{dt} [\cos 2t \ln t]$$

$$\Rightarrow \frac{f'(x)}{f(x)} = -2 \sin 2t \times \ln t + \cos 2t \times \frac{1}{t}$$

$$\Rightarrow f'(x) = f(x) \left[-2 \sin 2t \times \ln t + \cos 2t \times \frac{1}{t} \right]$$

$$19. f(x) = 2e^{1-x} \Rightarrow f'(x) = -2e^{1-x}$$

$$\Rightarrow f(1) = 2 \text{ and } f'(1) = -2$$

$$\text{The equation is } y - f(1) = f'(1)(x - 2)$$

$$\Rightarrow y - 1 = -2(x - 2)$$

$$\Rightarrow y = -2x + 5$$

$$20. f'(x) = \frac{d}{dx} \frac{\ln 2x}{x^3} = \frac{\frac{2}{2x} \times x^3 - \ln 2x \times 3x^2}{(x^3)^2} = \frac{x^2 - \ln 2x \times 3x^2}{x^6} = \frac{1 - 3 \ln 2x}{x^4}$$

$$f'(x) = 0 \Rightarrow 1 - 3 \ln(2x) = 0 \Rightarrow \ln(2x) = \frac{1}{3} \Rightarrow 2x = e^{1/3} \Rightarrow x = 0.5e^{1/3}$$

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